

A Self-Supervised Learning – Driven Method Empowering Multimodal Fatigue Assessment in Competitive Badminton

Lan Ding^a, Naichuan Zheng^b, Meng Liu^c, Weiyi Li^d,
Dapeng Bao^{e,*}, Hailun Xia^{b,*}

^a Department of Physical Education, Beijing University of Posts and Telecommunications, Beijing 100876, China

^b School of Information and Communication Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China

^c University of Jinan, Jinan 250000, China

^d International School, Beijing University of Posts and Telecommunications, Beijing 100876, China

^e Key Laboratory for Performance Training and Recovery of General Administration of Sport, Beijing Sport University, Beijing 100084, China

* Corresponding authors: Dapeng Bao, Email: baodp@bsu.edu.cn; Tel: +86 13911995047; Hailun Xia, Email: hailunxia@bupt.edu.cn

Abstract

Purpose: To develop and evaluate a self-supervised learning framework capable of continuously quantifying physical and mental fatigue in badminton athletes during high-intensity training, and to examine whether the derived fatigue index corresponds with physiological, perceptual, and sport-specific performance deterioration.

Methods: Twenty-four elite university badminton athletes completed a standardized fatigue-inducing protocol consisting of repeated countermovement jumps, sprints, and randomized hitting tasks designed to concurrently elicit neuromuscular and cognitive fatigue. Multimodal data—including HRV (RMSSD), countermovement jump height, IMU-based kinematics, Rating of Perceived Exertion (RPE), and Visual Analog Fatigue Scale (VAFS) scores—were collected across four fatigue stages (T0 – T60). Smash speed and accuracy served as weak supervisory signals. A self-supervised ranking model was trained using temporal progression and performance decline to generate a continuous fatigue index. Model outputs were evaluated using temporal consistency, correlations with physiological and perceptual indicators, and alignment with sport-specific performance decline.

Results: The fatigue index increased monotonically for all athletes (mean $\pm$ SD: 3.14 ± 1.40 at T0 to 82.98 ± 6.49 at T60), mirroring progressive neuromuscular deterioration (CMJ: -32.0%), reductions in HRV (-42.5% at T40), and substantial increases in mental and physical fatigue (VAFS: $+260.6\%$; RPE: $+7.21$). Smash speed (-10.6%) and accuracy (-46.3%) declined in parallel with increases in fatigue index, confirming strong performance-fatigue coupling. RMSSD deviation showed the most stable association with the fatigue index ($r = 0.18$), while kinematic disruptions further reflected deteriorating sensorimotor control.

Conclusions: The proposed label-free fatigue assessment approach effectively integrates multisource physiological and performance signals to produce a biologically interpretable fatigue index that tracks the multidimensional progression of fatigue under ecologically valid training conditions. Findings highlight the value of self-supervised learning for athlete monitoring and support its potential to inform individualized load management, technical – tactical planning, and fatigue-related injury prevention in competitive badminton.

Keywords: Self-supervised learning; Multimodal fatigue assessment; Neuromuscular and cognitive fatigue; Physiological signal representation; Athlete performance monitoring

1. Introduction

Badminton is a high-intensity, intermittent racket sport requiring repeated rapid changes of direction, jumps and overhead strokes, which imposes substantial physical and cognitive demands on athletes during competition^{1,2}. The forehand jump smash is a key attacking skill whose effectiveness depends on extreme shuttlecock speed, power output and precise timing, and is closely related to point-winning success^{3,5}. Experimental data further show that, after fatigue, jump smash shuttlecock speed and upper-limb kinematics are significantly impaired⁶, consistent with speed – accuracy trade-off and impulse-variability theory which predict decreased accuracy under high-force, high-speed, and fatigued conditions⁷. At the mechanistic level, both central and peripheral fatigue jointly contribute to this degradation of technical performance⁴.

Despite the widespread recognition of fatigue's importance, current assessment of fatigue status in badminton athletes still faces notable methodological limitations. On one hand, most studies isolate either physical fatigue or cognitive fatigue rather than examining their interaction. For example, research on neuromuscular fatigue such as Le Mansec et al.¹⁰ focuses solely on how peripheral muscle fatigue impairs smash speed and accuracy, while studies on mental fatigue such as Pageaux & Le Mansec⁹ and Kosack et al.¹⁴ emphasize slowed visuomotor responses and increased mental load without considering concurrent physical decline. Moreover, badminton-specific work on cognitive fatigue, including¹⁵, similarly examines attentional and decision-making fatigue independently. This fragmented approach fails to reflect the multidimensional and interactive nature of fatigue experienced during real competitive play.

On the other hand, traditional fatigue assessment methods also present practical challenges in real-sport environments. Subjective scales—commonly used in psychological fatigue studies such as Kosack et al.¹⁴—struggle to provide continuous, objective tracking of fatigue fluctuations during rally-based play. Laboratory-based biomechanical assessments, such as

the multi-marker motion capture systems used by Tong et al.¹¹ disrupt natural movement execution and frequently suffer from noise or data loss during high-intensity strokes and rapid directional changes. These limitations make it difficult to capture fatigue dynamically in ecologically valid contexts.

Therefore, achieving continuous, stable, and quantifiable fatigue assessment in real training and match environments remains a critical scientific challenge. Recent work demonstrates that competitive fatigue involves intertwined neuromuscular deterioration, perceptual-cognitive decline, and altered sensorimotor strategies—such as the changes in gaze behavior observed by Loiseau-Taupin et al.¹² and the progressive neuromuscular disruption documented by Lin et al.¹³—yet no existing framework integrates these multidimensional fatigue indicators into a unified monitoring approach.

Self-supervised learning (SSL) is a methodological framework that has shown potential advantages in this context¹⁶. Its core idea is to enable models to automatically extract physiologically meaningful intrinsic features from raw data through predefined learning tasks^{17,18}, without relying on extensive manual annotation. This approach is particularly suitable for research objects like fatigue that exhibit hidden, multimodal, and nonlinear characteristics^{19,20}, helping to uncover dynamic evolution patterns of physical and cognitive fatigue in the temporal dimension^{21,22}. Compared to traditional statistical methods, self-supervised learning can maintain higher feature representation stability in real-world contexts where noise, missing data, and natural fluctuations coexist^{23,24}, better aligning with the practical needs of sports training and competition monitoring.

Based on this, the present study aims to explore the application of intelligent models based on self-supervised learning in the assessment of "physical – mental" dual fatigue in badminton athletes. The study focuses on the following scientific questions: (1) How to learn robust feature representations sensitive to fatigue changes from multimodal data collected during real training; (2) How to construct a model framework capable of dynamically quantifying dual fatigue levels

and predicting degradation trends in key technical performance (such as smash speed and accuracy).

The significance of this study lies in promoting paradigm innovation in sports fatigue assessment methods. At the theoretical level, it reveals the interaction mechanisms between fatigue in movement control, power output, and cognitive processing through data-driven approaches, providing new evidence to support understanding of competitive state formation. At the practical level, this research is expected to construct an intelligent fatigue assessment solution with lower device dependency that can operate stably in real-world scenarios, providing scientific basis for personalized training load regulation, technical-tactical strategy optimization, and sports injury prevention, contributing to the advancement of scientific levels in competitive sports.

The significance of this study lies in promoting paradigm innovation in sports fatigue assessment methods. At the theoretical level, it reveals the interaction mechanisms between fatigue in movement control, power output, and cognitive processing through data-driven approaches, providing new evidence to support understanding of competitive state formation. At the practical level, this research is expected to construct an intelligent fatigue assessment solution with lower device dependency that can operate stably in real-world scenarios, providing scientific basis for personalized training load regulation, technical-tactical strategy optimization, and sports injury prevention, contributing to the advancement of scientific levels in competitive sports.

2. Methodology

2.1. Research Subjects, Methods, and Technical Route

2.1.1. Research Subjects

Sample size was determined using G*Power 3.1.9.7 (Heinrich Heine University, Germany) based on an F test (repeated-measures ANOVA, within factors), with an effect size $f = 0.30$, power = 0.80, and $\alpha = 0.05$, indicating a minimum requirement of 24 participants. Twenty-six elite university badminton athletes (first-class level or above) were initially recruited. After

screening with the University Student Learning Burnout Questionnaire and the Athlete Fatigue Questionnaire, two athletes showing signs of chronic psychological fatigue were excluded, leaving 24 participants for final analysis. The final sample consisted of 17 male athletes (age: 21.39 ± 1.91 years; height: 179.33 ± 6.25 cm; weight: 73.33 ± 7.24 kg) and 7 female athletes (age: 20.71 ± 0.95 years; height: 168.86 ± 6.49 cm; weight: 57.86 ± 3.93 kg). All participants were free from musculoskeletal injuries, voluntarily participated in the study, and provided written informed consent. The study protocol was approved by the Ethics Committee of Beijing Sport University (Approval No. 2023073H).

2.1.2. Research Methods

2.1.2.1. Experimental Method

This study adopted a single-factor repeated-measures experimental design, in which all measurements were collected repeatedly from the same participants at predefined time points during a continuous badminton-specific training protocol.

Data acquisition focused on time-dependent changes in neuromuscular performance, subjective fatigue perception, autonomic responses, and smash performance. Specifically, countermovement jump (CMJ) was used to assess neuromuscular performance, Rating of Perceived Exertion (RPE) and Visual Analog Fatigue Scale (VAFS) were used to capture subjective perceptions of physical and mental exertion, respectively, and heart rate (HR) signals were recorded to derive root mean square of successive differences (RMSSD) as an indicator of autonomic nervous system modulation. In addition, forehand smash speed and accuracy were measured to characterize time-dependent changes in sport-specific performance.

Each participant completed one experimental session lasting approximately 80 minutes. All tests were conducted in a standardized indoor facility combining a badminton court and laboratory space. Testing conditions were kept consistent across all participants.

(1) Experimental Design

The experiment followed a time-based repeated-measures design, with measurements collected at four predefined time points during a continuous fatigue-inducing protocol: baseline (T0), after 20 minutes

(T20), after 40 minutes (T40), and after 60 minutes (T60).

The purpose of this design was to examine how physiological, perceptual, and performance-related indicators evolve over time during prolonged high-intensity badminton activity.

A badminton-specific physical – mental load protocol was implemented, adapted from the fatigue protocol proposed by Le Mansec et al.¹⁴ (Figure 1). Each cycle consisted of three sequential tasks: ten countermovement jumps with hands on hips completed within 5 seconds; a 15-m sprint; ten randomized full-court hitting tasks.

Completion of these three tasks constituted one set. Ten sets formed one block, and two consecutive blocks constituted one time unit lasting approximately 20 minutes.

In addition to the physical load, sustained attentional demand was introduced through a randomized hitting task guided by visual signals. A badminton machine randomly delivered shuttlecocks to one of four predefined landing areas, while one of four target zones was simultaneously illuminated. Athletes were required to direct the shuttlecock to the illuminated zone, requiring continuous perception – decision – action coupling. This design ensured concurrent physical loading and cognitive engagement throughout the protocol.

Participants were verbally encouraged to maintain maximal effort during the entire protocol to ensure consistent task intensity across time.

(2) Experimental Procedure

Baseline measurement (T0): Participants first completed a standardized 10-minute warm-up, including jogging, muscle activation, dynamic stretching, neural activation, and movement integration. Following the warm-up, baseline data were collected. Participants performed CMJ testing, completed RPE and VAFS questionnaires, and then completed the first forehand smash speed and accuracy test. Heart rate signals recorded during this period were used to calculate baseline RMSSD.

Continuous task intervention: Participants then completed approximately 60 minutes of continuous badminton-specific activity, consisting of six blocks (Block 1 to Block 6). Every two consecutive blocks corresponded to one 20-minute time segment.

T20 measurement: Immediately after completion of Block 1 and Block 2, participants underwent forehand smash speed and accuracy testing, followed by CMJ testing and completion of RPE and VAFS questionnaires. Heart rate data collected during this segment were used to compute RMSSD.

T40 measurement: After completion of Block 3 and Block 4, the same measurement sequence was repeated: smash performance testing, CMJ assessment, subjective questionnaires, and RMSSD calculation.

T60 measurement: After completion of Block 5 and Block 6, participants again completed smash performance tests, CMJ testing, RPE and VAFS assessments, and RMSSD computation.

The overall experimental workflow is illustrated in Fig. 1.

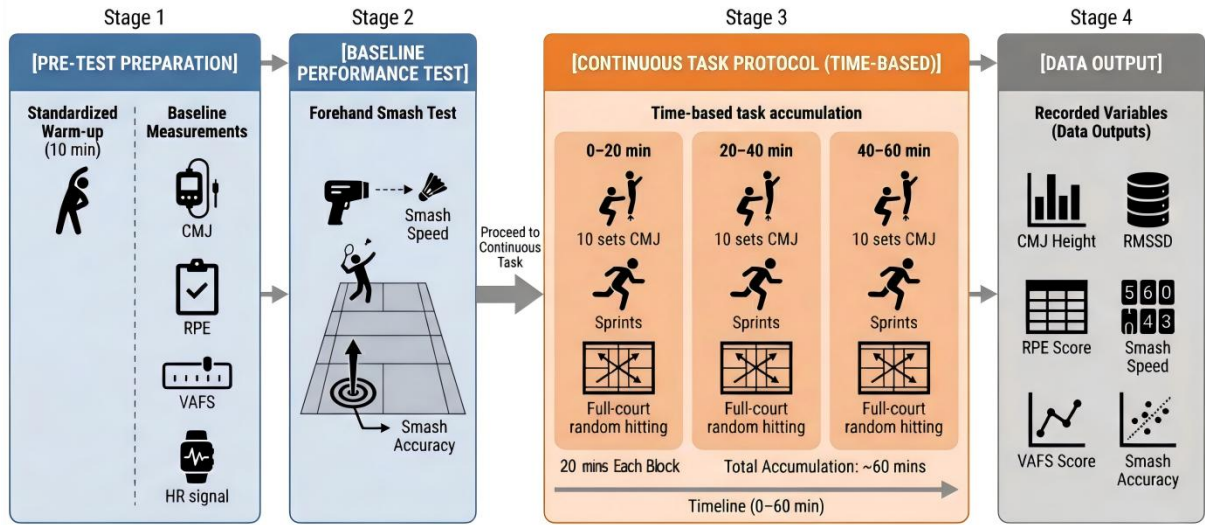


Fig. 1 Experimental procedure flowchart

(3) Measurement Indicators

At each predefined time point, fatigue- and performance-related outcomes were collected. Subjective fatigue was assessed using RPE (0 – 10) and the Visual Analog Fatigue Scale (VAFS, 0 – 9) to capture perceived physical exertion and mental strain, respectively; both were completed electronically immediately after each stage. Objective neuromuscular performance was evaluated using CMJ height. Sport-specific performance was assessed by forehand smash speed and accuracy.

(4) Test Content and Assessment Criteria

RPE and VAFS were administered at each time point based on participants' instantaneous subjective state. CMJ height was used to characterize time-dependent changes in neuromuscular performance. Participants performed a standardized countermovement followed by a maximal vertical jump, with three trials completed at each time point and the highest jump height retained for analysis. This CMJ assessment was conducted independently from the repeated jumps included in the load protocol, which served solely as part of the exercise load^{26,27}.

Fatigue progression across time points was described using typical ranges of change in CMJ, RPE, and VAFS, with physical fatigue reflected by declines in CMJ height and increases in RPE, and mental fatigue reflected by increases in VAFS scores²⁹.

Forehand smash performance was evaluated using a standardized test consisting of 10 maximal-effort forehand smashes to alternating target areas along the singles sideline. Smash landing accuracy was scored based on predefined target zones, and smash speed was recorded concurrently. Shuttlecock delivery was standardized using an automated feeding system to ensure consistent trajectory, speed, and release position³⁰. Detailed specifications of all experimental equipment are provided in the Supplementary Materials.

2.2. Self-supervised Learning Approach

This study adopts a self-supervised learning framework to derive a continuous fatigue index $F \in [0,100]$ from multi-source physiological and behavioral data collected from athletes. The fatigue index reflects fatigue intensity and maintains a monotonic relationship with sport-specific performance. Temporal order within each athlete serves as a weak supervisory signal, while Badminton speed and accuracy act as consistency references to ensure that increasing fatigue corresponds to performance decline.

During preprocessing, the variables RPE, VAFS, CMJ, and RMSSD are standardized. The CMJ values are inverted to obtain $\widehat{\text{CMJ}}$, where higher values

indicate lower fatigue. Because RMSSD may fluctuate non-monotonically across stages, its deviation from an individual baseline is calculated to quantify autonomic imbalance:

$$z = \frac{\text{RMSSD} - b}{s}, \quad d = |z|, \quad \text{sign}(z) \in \{-1, 0, 1\},$$

where b denotes the individual baseline (a long-term mean), and s represents the dispersion parameter such as the standard deviation or the median absolute deviation (MAD). Here, d captures the magnitude of deviation from autonomic steady state, and $\text{sign}(z)$ indicates the direction of deviation (-1 for lower-side, 1 for higher-side, and 0 for no meaningful change).

The final feature vector is

$$\tilde{\mathbf{x}} = [\text{RPE}, \text{VAFS}, \widehat{\text{CMJ}}, d, \text{sign}(z)].$$

A multilayer perceptron (MLP) $F_\theta: \mathbb{R}^5 \rightarrow \mathbb{R}$ with parameter set θ maps each feature vector to a raw fatigue score $F_\theta(\tilde{\mathbf{x}})$, as shown in Fig. 2. To preserve physiological interpretability, monotonicity constraints are imposed so that increases in RPE, VAFS, or physiological deviation do not reduce the predicted fatigue index.

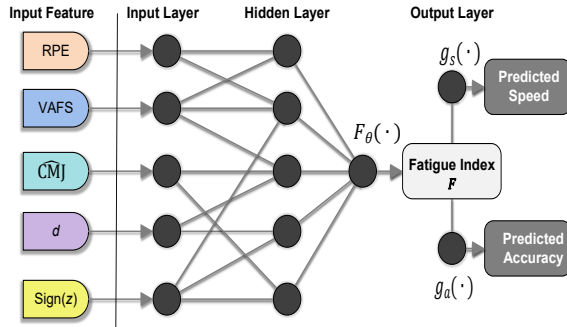


Fig. 2 The MLP model

Model training follows a self-supervised ranking paradigm. Within each athlete, ordered observation pairs ($i < j$) are constructed to encourage the model to assign higher fatigue scores to later or more fatigued states:

$$\mathcal{L}_{\text{rank-time}} = \sum_{(i < j) \in F_\theta(\tilde{\mathbf{x}}_i)} \log(1 + \exp(-[F_\theta(\tilde{\mathbf{x}}_j) - F_\theta(\tilde{\mathbf{x}}_i)])).$$

To reinforce consistency with performance, the ranking loss is weighted by the relative decline in Badminton speed and accuracy, which are normalized by their interquartile ranges to minimize the effect of outliers. In addition, monotone decoding functions

$g_s(\cdot)$ and $g_a(\cdot)$ predict Badminton speed and accuracy from the fatigue index, with the Huber loss quantifying prediction errors:

$$\mathcal{L}_{\text{iso}} = \text{Huber}(g_s(F_\theta), \text{Speed}) + \text{Huber}(g_a(F_\theta), \text{Acc})$$

The total training objective combines the temporal ranking loss, the performance-consistency constraint, the monotone decoding loss, and a regularization term:

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda_{\text{iso}} \mathcal{L}_{\text{iso}} + \lambda \|\theta\|_2^2.$$

In this formulation, λ_{iso} controls the contribution of the performance-decoding term, balancing the consistency between the fatigue index and task performance, while λ is the coefficient of the regularization that prevents overfitting by penalizing large parameter magnitudes. Both coefficients are tuned empirically to achieve the best trade-off between model accuracy and generalization.

After training, the raw fatigue scores are linearly mapped to the range $[0, 100]$ through robust calibration to obtain the normalized fatigue index F . The model is optimized using the Adam algorithm with a learning rate of 0.001, a batch size of 8, and early stopping when no improvement is observed for 20 consecutive epochs.

3. Experimental Results and Analysis

This section evaluates the effectiveness of the proposed fatigue-index model through detailed analyses of the experimental data and examines the relationships between the independent variables, weak supervisory signals, and the fatigue index. By analyzing temporal trajectories, correlations, and the stability of model outputs, we verify that the model, under unlabeled conditions, captures meaningful variations in fatigue.

3.1. Temporal evolution of independent variables and weak supervisory signals

To ground the subsequent model evaluation in empirical evidence, this subsection characterizes how the input features and the weak supervisory signals evolve over time.

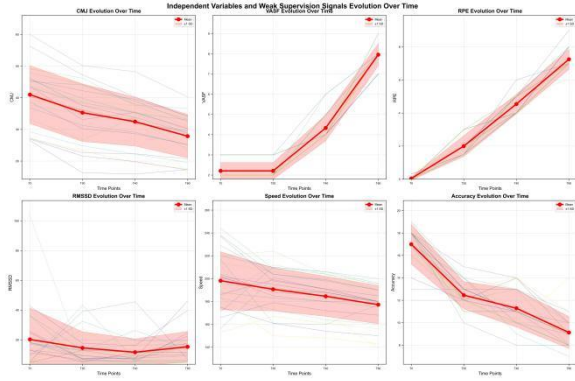


Fig. 2 Temporal evolution of independent variables (CMJ, VAFS, RPE, RMSSD) and weak supervisory signals (Speed, Accuracy) across T0 - T60.

Fig. 3 synthesizes the temporal trajectories from T0 to T60 and provides the visual context for Table 1.

Among the independent variables, CMJ shows a non-linear decline with the steepest drop between T40 and T60, consistent with progressive neuromuscular fatigue. VAFS and RPE rise monotonically; VAFS accelerates after T40, whereas RPE increases more steadily with a large early gain from a near-zero baseline. RMSSD is plotted as raw HRV to aid interpretation and displays a decrease through T40 with partial rebound by T60 the broad baseline spread highlights inter-individual variability and motivates the deviation-from-baseline representation used in the model.

Across T0 - T60, the independent variables show a coherent fatigue progression. CMJ declines from 41.001 ± 9.161 cm at T0 to 27.862 ± 6.859 cm at T60 (total $\Delta = -13.139$, -32.0%), with the largest

interval decrease occurring from T40 to T60. VAFS increases by $+5.750$ overall ($+260.6\%$), with the main rises occurring during T20→T40 and T40→T60. RPE increases by $+7.213$ across the session; because baseline RPE is near zero, percent change is inflated and absolute changes are more informative. RMSSD decreases to 11.647 ± 9.006 ms at T40 (-42.5% vs. T0) and then partially rebounds to 15.327 ± 10.326 ms at T60 ($+3.681$ vs. T40).

Consistent with the temporal deterioration observed in Fig. 3, weak supervisory signals also decline over time. Speed decreases approximately linearly from 198.208 ± 25.461 to 177.208 ± 16.647 (total $\Delta = -21.000$, -10.6%). Accuracy shows a larger relative reduction from 17.000 ± 1.769 to 9.125 ± 1.424 (total $\Delta = -7.875$, -46.3%), with the steepest drop from T0 to T20 and continued decline thereafter. Together, convergent trends across neuromuscular (CMJ), subjective (VAFS/RPE), autonomic (RMSSD), and performance measures (Speed/Accuracy) support progressive fatigue and motivate the model design choices (inverted CMJ, baseline-centered RMSSD deviation, and monotonic consistency between the fatigue index and both subjective scales and sport-specific performance). Detailed descriptive statistics (mean $\pm$ SD), absolute changes (Δ), and percent changes across time points are provided in the Supplementary Materials.

3.2. Fatigue index evolution over time

Building on the feature- and performance-level trends, this subsection characterizes how the model-derived fatigue index F (scaled to $[0,100]$) evolves across the session.

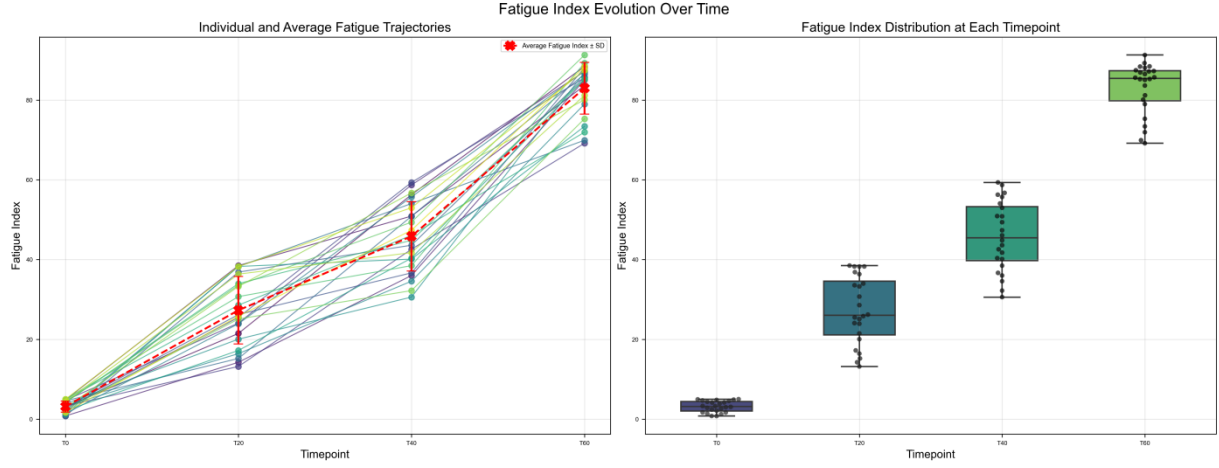


Fig. 3 Fatigue Index Evolution and Distribution Across Timepoints

displays individual trajectories with the group mean $\pm$ SD (left) and the distribution at each time point (right). The trajectories show a strictly monotonic increase for all participants, and the group

mean rises steadily from early to late stages, indicating progressive fatigue accumulation captured by the model under unlabeled conditions.

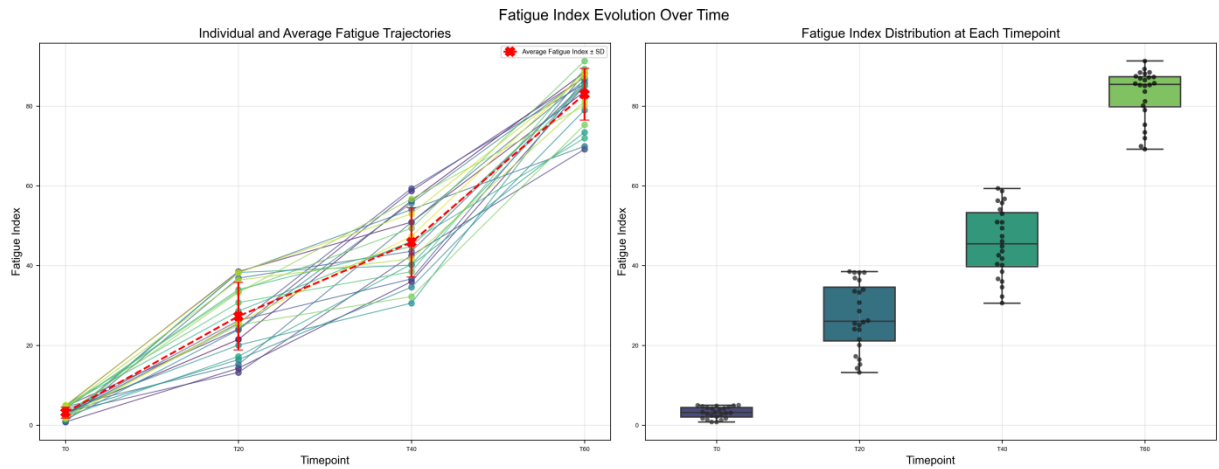


Fig. 3 Fatigue Index Evolution and Distribution Across Timepoints

Quantitatively, the group mean ($\pm$ SD) progresses from 3.141 ± 1.404 at T0 to 27.296 ± 8.464 at T20, 45.841 ± 8.680 at T40, and 82.983 ± 6.492 at T60. Interval-wise changes confirm an early surge, a mid-phase plateauing, and a late acceleration: the average increase is $+24.155 \pm 8.417$ from T0 $\rightarrow$ T20, $+18.545 \pm 10.429$ from T20 $\rightarrow$ T40, and $+37.142 \pm 10.005$ from T40 $\rightarrow$ T60, yielding a total gain of $+79.842 \pm 6.409$ (range 65.933 – 87.141). Dispersion

expands from a low-variability baseline ($SD=1.404$) to a mid-session peak ($SD \approx 8.5 - 8.7$ at T20 – T40), then contracts by T60 ($SD=6.492$), as also reflected by the timepoint boxplots. Together, these patterns—strictly increasing individual paths, stage-dependent growth, and the modulation of variability—are coherent with the observed declines in Speed and Accuracy and with the neuromuscular/autonomic signals (CMJ, RMSSD). They support the model's monotonic design and

indicate that F provides a sensitive and interpretable summary of fatigue progression over time.

3.3. Correlation and Feature Interaction Analysis

To verify the physiological plausibility and internal consistency of the proposed fatigue index under multi-source input conditions, correlations between the input features (CMJ, VAFS, RPE, RMSSD deviation), weak supervisory signals (stroke speed and accuracy), and the fatigue index F were examined. Pearson correlation coefficients (r), statistical significance (p -values), and effect sizes (Cohen's d) were calculated at each time point (T0, T20, T40, T60).

Overall, correlations between individual features and the fatigue index were generally weak to moderate, consistent with the model design that integrates multiple heterogeneous signals rather than relying on any single indicator. Among all variables, RMSSD deviation exhibited the strongest and most consistent association with the fatigue index (average $r = 0.179$), peaking at T0 ($r = 0.333$) and remaining elevated at T60 ($r = 0.291$), with a very large effect size ($d = 4.676$). This pattern indicates that baseline-centered HRV deviation is a sensitive marker of autonomic fatigue accumulation.

CMJ demonstrated a persistent negative correlation with the fatigue index (average $r = -0.124$, $d = 3.660$),

with the strongest association observed at T40 ($r = -0.204$), reflecting progressive neuromuscular performance decline under fatigue. RPE and VAFS showed weaker mean correlations with the fatigue index (average $r = 0.065$ and -0.074 , respectively), which may be attributable to subjective scale saturation effects; nevertheless, their temporal increases remained directionally consistent with fatigue accumulation.

Performance-related weak supervisory signals showed significant negative associations with fatigue in the later stages of the protocol. Stroke speed exhibited its strongest negative correlation at T60 ($r = -0.476$, $p = 0.019$, $d = 3.574$), while accuracy showed the largest decline at T40 ($r = -0.429$, $p = 0.036$, $d = 3.956$). These results indicate that accumulated fatigue not only reduces movement speed but also impairs task precision.

Taken together, the consistent directionality and effect sizes across neuromuscular (CMJ), subjective (RPE, VAFS), autonomic (RMSSD deviation), and performance (speed, accuracy) variables provide convergent statistical support for the validity of the proposed fatigue index. The observed correlations confirm that increases in the fatigue index are reliably associated with physiological deterioration and performance decrements across time.

Table 1 Pearson Correlations (r), Statistical Significance (p -values) and Effect Size(d) between independent variables and weak supervisory signals and the fatigue index across T0 - T60.

Measure	Avg r	Max r (time)	Min r (time)	r (T0, T20, T40, T60)	p-values (T0, T20, T40, T60)	Effect Size (d)
CMJ	-0.124	-0.072 (T20)	-0.204 (T40)	[-0.127, -0.072, -0.204, -0.092]	[0.556, 0.736, 0.338, 0.668]	3.660
VAFS	-0.074	0.071 (T0)	-0.241 (T20)	[0.071, -0.241, -0.018, -0.108]	[0.741, 0.256, 0.933, 0.617]	2.411
RMSSD Dev.	0.179	0.333 (T0)	0.043 (T20)	[0.333, 0.043, 0.050, 0.291]	[0.112, 0.842, 0.818, 0.168]	4.676
RPE	0.065	0.118 (T60)	0.004 (T40)	[0.046, 0.094, 0.004, 0.118]	[0.830, 0.663, 0.987, 0.583]	2.960
Speed	-0.163	0.087 (T20)	-0.476 (T60)	[-0.257, 0.087, -0.008, -0.476]	[0.226, 0.686, 0.972, 0.019]	3.574
Accuracy	-0.114	0.166 (T60)	-0.429 (T40)	[-0.043, -0.149, -0.429, 0.166]	[0.841, 0.488, 0.036, 0.438]	3.956

^A Correlation r quantifies the linear relationship strength with the fatigue index, significance (p-value) indicates statistical reliability, and effect size (Cohen's d) reflects the practical magnitude of variable influence.

4. Discussion

4.1. Summary and Interpretation of Key Findings

This study developed a self-supervised learning – based model capable of continuously quantifying both physical and mental fatigue in badminton athletes without relying on manually annotated labels. The findings demonstrate that the proposed model preserves a stable monotonic trajectory under multisource data fluctuations and exhibits strong consistency with sport-specific performance indicators, namely smash speed and accuracy. As training progressed, the fatigue index F showed a strictly increasing pattern that aligned closely with objective neuromuscular measures (declines in CMJ height), autonomic nervous system responses (baseline deviation of HRV – RMSSD), and subjective perceptions of fatigue (RPE, VAFS). These

results suggest that weak supervisory constraints — temporal ordering and performance deterioration — are sufficient for the model to learn physiologically meaningful fatigue representations.

This pattern corresponds well with the multidimensional fatigue accumulation mechanisms reported in prior literature. In high-intensity intermittent sports such as badminton, fatigue typically involves concurrent central fatigue, peripheral neuromuscular fatigue, and decrements in visuomotor processing^{32,33,34}. The linear decline in smash speed and accuracy observed in this study is also consistent with findings by Le Mansec et al.³⁵, who showed that muscular fatigue markedly impairs power generation, movement control, and temporal precision during the jump smash. Furthermore, the decline in accuracy aligns with the speed – accuracy trade-off framework³⁶, indicating that maintaining high force output under fatigue comes at the cost of reduced spatial control and shot placement precision.

4.2. Multi-Dimensional Mechanisms Underlying the Model Behavior

From a mechanistic perspective, the present findings further support the theoretical framework of physical – cognitive coupled fatigue. Numerous studies have demonstrated that in racket sports requiring rapid visual processing and time-constrained decision-making — such as badminton and table tennis — mental fatigue can substantially impair reaction time, movement accuracy, and gaze behavior^{37,38}. These effects were clearly reflected in our dataset.

Both RPE and VAFS exhibited accelerated increases, especially from T40 to T60, suggesting that mental fatigue is amplified when superimposed on high levels of physical fatigue³⁹. Additionally, HRV baseline deviation showed a persistent positive association with the fatigue index ($r \approx 0.18$), indicating that autonomic imbalance constitutes a core physiological mechanism underlying cumulative fatigue⁴⁰.

Our modeling of RMSSD deviation relative to individual baselines aligns with sport physiology research highlighting the importance of individualized HRV profiling. Prior studies have shown that baseline-centered HRV is more effective for identifying fatigue than absolute HRV values^{41,42}. The present results provide algorithmic evidence supporting this physiological rationale.

4.3. Advantages of the Proposed Self-Supervised Approach

Compared with traditional fatigue assessment approaches, the proposed self-supervised framework exhibits several distinct advantages:

(1) Label-free and robust in real-world environments

Conventional fatigue evaluation relies heavily on annotated data—such as athlete self-reports or coach ratings—which are subjective, labor-intensive, and limited in temporal resolution. By leveraging temporal ordering and consistency with sport-specific performance as weak supervisory signals, the proposed model autonomously extracts physiologically meaningful features from raw IMU,

HRV, and CMJ data. Similar advantages of weak-label-driven representation learning have been demonstrated in clinical time-series modeling⁴³ and large-scale wearable sensing applications⁴⁴.

(2) Integration of heterogeneous multimodal data reflecting the multidimensional nature of fatigue

Traditional research frequently isolates physical fatigue from cognitive fatigue⁴⁵, overlooking their interaction during real competition. In contrast, the present integrated framework effectively captures:

- the synchronized decline in neuromuscular function and deviation in RMSSD;
- the coupling between mental fatigue and reductions in movement precision;
- the strictly monotonic within-individual fatigue trajectories.

These results highlight the suitability of self-supervised learning for handling the complexity, multimodality, and nonlinearity inherent in fatigue-related time-series data^{46,47}.

4.4. Limitations and Future Work

Despite its promising results, the present study has several limitations:

(1) Sample size and scope of sport discipline

The model was trained on 24 university-level badminton athletes, and although internal consistency was strong, external generalizability requires further validation. Future research should include athletes of different competition levels, age groups, and sport specialties, and compare findings with other high-intensity sports such as tennis or basketball.

(2) Limited inclusion of physiological modalities

The present study did not incorporate EMG, blood lactate, EEG, or other physiological signals that more directly reflect neuromuscular or metabolic fatigue. Previous research has demonstrated that EMG can sensitively detect changes in muscle activation and fatigue-induced alterations⁴⁸, suggesting that its integration may enhance the physiological interpretability of the fatigue index.

(3) Model interpretability remains limited

Although the monotonic fatigue trajectories reflect strong physiological plausibility, the self-supervised model still functions largely as a black box. Future work could incorporate explainable AI techniques—

such as SHAP values, attention visualization, or prototype-based interpretability — to improve transparency and provide coaches and practitioners with clearer insights into the decision-making process of the model.

4.5. Practical Implications for Athlete Monitoring

The proposed approach has important implications for applied training and fatigue management:

Real-time monitoring of fatigue trajectories: The fatigue index provides a sensitive depiction of fatigue accumulation, enabling coaches to dynamically adjust training loads.

Optimization of technique training under fatigue: Notable decreases in smash accuracy during periods of accelerated fatigue (e.g., T40 – T60) suggest that high-precision technical training should be avoided during these stages.

Injury prevention and overtraining management: Coupled monitoring of HRV deviation and CMJ decline can facilitate identification of high-risk periods where cumulative central and peripheral fatigue increases injury susceptibility.

Support for individualized training programs: By capturing within-athlete fatigue dynamics, the model offers quantitative evidence for personalized load management.

These implications align closely with global trends toward intelligent, data-driven athlete monitoring and performance optimization.

5. Conclusion

This study presents a novel self-supervised learning framework capable of continuously quantifying both physical and mental fatigue in badminton athletes without dependence on manually annotated labels. By integrating multisource physiological, kinematic, and performance data, the proposed model yielded a robust monotonic fatigue index that aligned with neuromuscular decline, autonomic imbalance, subjective fatigue perception, and sport-specific performance deterioration during prolonged high-intensity training. These findings demonstrate that weak supervisory cues — temporal progression and performance consistency — are sufficient for deriving

physiologically meaningful fatigue representations in ecologically valid contexts.

The proposed approach advances fatigue assessment beyond traditional methods by capturing the multidimensional and interacting nature of physical and cognitive fatigue and by offering a scalable, interpretable, and low-burden solution for real-time monitoring. Practically, the fatigue index provides coaches and practitioners with a sensitive tool for managing training load, mitigating performance decrements, and reducing injury risk in high-performance environments.

This study did not perform sex-based analyses, which may limit the generalizability of the findings. Future work should extend this framework to larger and more diverse athlete populations, incorporate additional physiological modalities such as EMG or metabolic markers, and explore model interpretability to enhance its applicability in applied sport settings. Taken together, this study contributes an important methodological step toward data-driven, individualized fatigue management in competitive sport.

Acknowledgments

The authors would like to express their sincere gratitude to Beijing Sport University, particularly the Key Laboratory for Performance Training and Recovery of the General Administration of Sport, for providing the experimental venues, equipment, and access to participant recruitment.

The substantial support from these institutions was essential for the successful completion of data collection and experimental procedures. This study was supported by the “Support Program” of the Department of Physical Education, Beijing University of Posts and Telecommunications (2025–2026).

Authors’ contributions

Lan Ding conceived the study, designed the overall experimental protocol, and was responsible for data collection. Naichuan Zheng performed the experimental data analysis and drafted the manuscript. Meng Liu participated in data collection. Weiyi Li was

responsible for manuscript formatting and layout. Dapeng Bao and Hailun Xia served as corresponding authors, supervised the study, and critically revised the manuscript. All authors read and approved the final manuscript and agreed to the order of authorship.

Competing interests

The authors declare that they have no competing interests.

Declaration of generative AI in scientific writing

During the preparation of this work, the authors used ChatGPT to improve language clarity and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Reference

- Pardiwala DN, et al. Badminton injuries in elite athletes: a review of epidemiology and biomechanics. *Indian J Orthop.* 2020;54(3):237-245.
- Phomsoupha M, Laffaye G. Injuries in badminton: A review. *Sci Sports.* 2020;35(4):189-199.
- Rusdiana A, et al. Effect of fatigue on biomechanical variable changes in overhead badminton jump smash. *Ann. Appl. Sport Sci.* 2020;8(3):0-0.
- Tornero-Aguilera JF, Jimenez-Morcillo J, Rubio-Zarapuz A, Clemente-Suárez VJ. Central and peripheral fatigue in physical exercise explained: A narrative review. *Int J Environ Res Public Health.* 2022;19(7):3909.
- King M, Towler H, Dillon R, McErlain-Naylor S. A correlational analysis of shuttlecock speed kinematic determinants in the badminton jump smash. *Appl Sci.* 2020;10(4):1248.
- Rusdiana A. 3D kinematics analysis of overhead backhand and forehand smash techniques in badminton. *Ann Appl Sport Sci.* 2021;9(3):e1002.
- Molina SL, Bott TS, Stodden DF. Applications of the Speed–Accuracy Trade-off and Impulse-Variability Theory for Teaching Ballistic Motor Skills. *J Mot Behav.* 2019; in press.
- Le Mansec Y, Perez J, Rouault Q, Doron J, Jubeau M. Impaired performance of the smash stroke in badminton induced by muscle fatigue. *Int J Sports Physiol Perform.* 2020;15(1):52-59. doi:10.1123/ijsp.2018-0697.
- Le Mansec Y, Pageaux B, Nordez A, Dorel S, Jubeau M. Mental fatigue alters the speed and the accuracy of the ball in table tennis. *J Sports Sci.* 2018;36(23):2751-2759. doi:10.1080/02640414.2017.1418647.
- Le Mansec Y, Perez J, Rouault Q, Doron J, Jubeau M. Impaired performance of the smash stroke in badminton induced by muscle fatigue. *Int J Sports Physiol Perform.* 2020;15(1):52-59. doi:10.1123/ijsp.2018-0697.
- Tong J, et al. The effects of ankle dorsiflexor fatigue on lower limb biomechanics during badminton forward forehand and backhand lunge. *Front Bioeng Biotechnol.* 2023;11:1013100.
- Loiseau-Taupin M, et al. Effects of acute physical fatigue on gaze behavior and performance during a badminton game. *Front Sports Act Living.* 2021;3:725625.
- Lin Z, Blazevich AJ, Abbiss CR, Cochrane Wilkie J, Nosaka K. Neuromuscular fatigue and muscle damage following a simulated singles badminton match. *Eur J Appl Physiol.* 2023;123(6):1229-1240. doi:10.1007/s00421-023-05148-w.
- Kosack MH, et al. The acute effect of mental fatigue on badminton performance in elite players. *Int J Sports Physiol Perform.* 2020;15(5):632-638.
- Van Cutsem J, et al. Mental fatigue impairs visuomotor response time in badminton players and controls. *Psychol Sport Exerc.* 2019;45:101579.
- Yuan H, Zhang Y, Sun J, et al. Self-supervised learning for human activity recognition using 700,000 person-days of wearable data. *Nat Commun.* 2024;15:3283.
- Liu Z, Alavi A, Li M, Zhang X. Self-Supervised Contrastive Learning for Medical Time Series: A Systematic Review. *Sensors (Basel).* 2023;23(9):4221.
- Raghu A, Chandak P, Alam R, Guttig J, Stultz C. Sequential Multi-Dimensional Self-Supervised Learning for Clinical Time Series. *Proc Mach Learn Res.* 2023;202:4602–17.
- Hallgarten P, Bethge D, Özdenizci O, et al. TS-MoCo: Time-Series Momentum Contrast for Self-Supervised Physiological Representation Learning. *Proc Eur Signal Process Conf.* 2023:1030–4.
- Chen H, Lundberg SM, Erion G, Kim JH, Lee SI. Forecasting adverse surgical events using self-supervised transfer learning for physiological signals. *npj Digit Med.* 2021;4:164.
- Deng S, Teng D, Guo Z, et al. Self-Supervised Federated Learning for Personalized Human Activity Recognition. *Proc AAAI Conf Artif Intell.* 2024;38(13):14285–93.
- Small SR, Kelleher S, O'Brien J, et al. Self-Supervised Machine Learning to Characterize Step Parameters from Wearable Sensor Data. *Sensors (Basel).* 2024;24(10):3198.
- Cañellas ML, Väänänen H, Alasaarela E. A self-supervised multimodal framework for 1D physiological data fusion in remote health monitoring. *Inf Fusion.* 2025;110:103397.
- Eldele K, Chen Z, Wu M, et al. Self-supervised representation learning for time-series data: A systematic review. *IEEE Trans Neural Netw Learn Syst.* 2023;34(12):10397–410.
- De Vries J, Michielsen H, Van Heck GL, Drent M. Measuring fatigue in sarcoidosis: the Fatigue Assessment Scale (FAS). *Br J Health Psychol.* 2004;9(3):279-91.
- Borg GA. Psychophysical bases of perceived exertion. *Med Sci Sports Exerc.* 1982;14(5):377-81.